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B.Tech in Electronics and Communication Engineering (IoT)
Sep 2023 – July 2027
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Education

Madan Mohan Malaviya University of Technology, Gorakhpur — Sep 2023 – July 2027
B.Tech in Electronics and Communication Engineering (IoT) — *CGPA: 8.10/10.0*

Projects

AgenticForge – Multi-Agent AI Software Engineering Platform — May 2026 – Present
LangGraph, LangChain, Google Gemini API, Node.js, Express.js, Redis, Docker, WebSockets, React

- Architected a multi-agent AI platform with 8 specialized LLM agents (PM, Architect, Planner, Coder, Reviewer, Executor, Debugger, Deploy) orchestrated through a cyclical LangGraph workflow to automate the end-to-end software development lifecycle.
- Engineered Redis-backed state checkpointing for 8-agent workflows, enabling recovery from failed nodes and eliminating full workflow restarts after crashes.
- Integrated the Google Gemini API with a token-budgeting layer to monitor per-call costs across 8 autonomous agents and automatically halt execution when configurable spending limits were exceeded.
- Built a Docker-based execution sandbox with Git-backed rollbacks to compile, test, and validate AI-generated code in isolated containers, automatically reverting failed builds to the last stable version.
- Developed a real-time React dashboard with an Express REST API and WebSocket server, streaming live logs, token usage, and build status across 8 AI agents.

CodeJudge | *Node.js, Express.js, MongoDB, JWT, Judge0 API* — Feb 2026 – Apr 2026

- Built a full-stack online coding judge using MongoDB, Express.js, React, and Node.js (MERN), enabling users to solve coding problems, execute code, and submit solutions for automated evaluation.
- Integrated the Judge0 API to support real-time compilation and execution across multiple programming languages, generating automated verdicts including Accepted, Wrong Answer, and Runtime Error.
- Implemented JWT-based authentication and role-based access control (RBAC) with separate admin and user roles, securing problem management, submissions, and protected routes.
- Designed a problem and submission management system supporting structured test cases, constraints, input/output schemas, and AI-assisted hints for debugging and learning support.

Technical Skills

Languages: C++, JavaScript (ES6+)
Frontend: React.js, Redux Toolkit, JavaScript, HTML5, CSS3, Tailwind CSS, React Router
Backend: Node.js, Express.js, REST APIs, JWT Authentication, WebSockets
Databases: MongoDB, Mongoose, Redis
AI / GenAI: LLM Integration, Google Gemini API, AI Agents & Multi-Agent Systems, LangGraph, Prompt Engineering
Developer Tools: Git, GitHub, Docker, VS Code, Postman, npm
Core CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, REST API Design, System Design (Fundamentals)

Achievements

- Qualified for **Round 2 of TCS CodeVita 2025**, a global competitive programming contest, demonstrating strong skills in **Data Structures, Algorithms, and Problem Solving**.
- **Runner-up**, State-Level Kho-Kho Tournament — reflecting teamwork, discipline, and a competitive mindset